



BOUAROUR Ayoub

Software engineer | Agentic Systems Developer

📍 Oujda, Morocco ✉ bouarourayoub0@gmail.com ☎ +212627049264

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PROFILE

AI Software Engineer specializing in the design and development of agentic AI systems, from LLM-driven workflows to autonomous AI agents and multi-agent architectures (MAS). Strong experience in prompt engineering and context engineering to improve agent reasoning, reliability, and output quality.

Skilled in connecting AI agents to legacy and enterprise systems, grounding them with rich contextual data via MCP, and deploying them into real-world applications. Experienced in integrating agents into client products using AG-UI and CopilotKit to enable seamless human-AI interaction.

Driven by the belief that software is evolving toward a new paradigm where LLMs act as operating systems and natural language becomes the primary programming interface

PROFESSIONAL EXPERIENCE

Software Engineer, OpenSNZ-Technology

2025/09 – Present

Software Engineer and AI Developer specializing in the design, implementation, and maintenance of agentic AI systems, from rapid prototyping to production-ready solutions, including complex agent-driven architectures.

Skills : Agentic AI, Context Engineering, Prompt Engineering, MCP, LangChain/LangGraph, LangSmith, RAG system, CopilotKit(AG-UI)

PFE Internship | AI Agent Developer,

OpenSNZ-Technology

2025/02 – 2025/08 | oujda, Morocco

Developed a Multi-Agent System for intelligent pharmaceutical catalog analysis, automating data extraction from raw files and coordinating specialized AI agents for querying, cost aggregation, reporting, and decision support. Designed with a modular, scalable, and maintainable agent-based architecture.

Skills : LLM-driven Workflows, AI agents, Multi-Agent Systems, LangChain/LangGraph, LangSmith, NextJS, Python, Knowledge Graph, Prompt Engineering, Context Engineering, RAG, MCP, CopilotKit(AG-UI)

PROJECTS

Brikool, A platform designed to help local freelancers list and showcase their services, while providing customers with easy access and fast discovery of service providers

Skills : Next.js, Auth.js, Typescript, Spring, Spring Boot, Tailwind CSS, PostgreSQL, Docker, Git, GitHub, Next-intl, Cypress, Docker, CI-CD, Jira, ShadcnUI

TECHNICAL SKILLS

Programming languages

Python, Typescript, Javascript, Java

AI architectures and principles

Context Engineering, Prompt Engineering, RAG Systems, AI Agents, Multi-Agent Systems, AI-driven Workflows

AI agent stack

LangChain/LangGraph, LangSmith (observability), CopilotKit, MCP protocol, AG-UI (Copilotkit) Protocol

WEB development

FastAPI, Next.js, ReactJS, Redux Toolkit, Zustand, ContextAPI, TailwindCSS, Chacn-UI, Radix-UI, Sass, Zod schema, Drizzle ORM, Auth.js, KindeAuth, Cypress, Next-intl

Databases

Neo4j, Postgres, Mysql, Redis, MongoDB

DevOps

Git, GitHub, Gitlab, Docker, Docker-compose, Github actions, Jira, CI/CD, Prometheus, Jeager, Grafana, Otel

Operating systems

Linux/Unix

EDUCATION

Specialized Master in Computer Science

Engineering, Faculty of Science (FSO)

2023 – 2025 | Oujda, Morocco

Bachelor's Degree in Mathematics and Computer

Sciences, Faculty of Sciences (FSO)

2019 – 2023 | Oujda, Morocco

ARAPT ☑, *Arabic Part-of-Speech (POS) Tagging System Using Deep Neural Networks*

Skills : ML/DL, NLP, BiLSTM, Keras, Python, Next.js, FastAPI, ReactJs, Git, Github, Shadcn, Next-intl

Mandoc-OCR ☑, *Optical Character Recognition (OCR) system for Arabic Hand writings*

Skills : ML/DL, Image Processing, CNN, Keras, Python, Nextjs, Git, Github, Github actions, CI-CD, Docker, Shadcn, Next-intl

UNI-Lab ☑, *A centralized platform for managing faculty laboratories and equipment, including material inventory and resource allocation*

Skills : Next.js, NextAuth, Spring, Spring Boot, Tailwind CSS, MySQL, Docker, Git, GitHub, DaisyUI

LANGUAGES

English (Proficient) | **French** (Technical) | **Arabic** (Native)

Scientific Baccalaureate (Physical Sciences),
Essaadiyine Qualifying High School

2019 | El Aioun Sidi Mellouk, Morocco

SOFT SKILLS

Future-Focused Vision

Analytical Thinking

Team Collaboration

Active Listening

Adaptability

INTERESTS

- Podcasts
- CTF Challenges
- Open Source
- Video Gaming
- Continuous Learning